

John Baik

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Education

University of Maryland

BS in Computer Science and Statistics

Expected: May 2026

- GPA: 3.51/4.0
- **Coursework:** Intro to Machine Learning, Intro to Data Science Algorithms, Intro to Probability Theory, Linear Algebra, Applied Probability and Statistics, Advanced Data Structures

Experience

Research Intern

National Cancer Institute

Bethesda, MD

June 2024 – Aug 2024

- **Designed and developed a neural network using PyTorch to classify pancreatic cell types (alpha, beta, ductal, acinar), using DNA transcription sequence data from genomes.**
- Engineered and preprocessed a large-scale biological dataset using Pandas and NumPy, implementing data normalization, feature extraction, and dimensionality reduction to improve model performance.
- Trained, validated, and optimized a custom neural network, incorporating tensor operations, loss function tuning, and scikit-learn metrics for a diverse performance evaluation to account for multiple conditions.
- Collaborated with cancer genomics researchers to translate biological questions into machine learning problems

Projects

Phishing Detector

[Phishing-Classifier](#)

- Developed a **classification model** to accurately detect phishing emails using real-world datasets and supervised learning techniques
- **Achieved 98% accuracy, 99% precision, and 99% recall**, displaying high accuracy in distinguishing phishing scams from legitimate emails
- Utilized tools including Jupyter Notebook, Pandas, NumPy, scikit-learn, and Matplotlib for data preprocessing, model training, evaluation, and visualization

Diabetes Classifier and EDA Project

[Diabetes-classifier](#)

- **Cleaned and standardized a patient biomarker dataset** by handling missing values, encoding categorical variables, and scaling features.
- **Ingested and managed the dataset using Snowflake**, uploading cleaned CSV files into cloud tables for structured querying and reproducible analysis.
- Performed comprehensive **EDA using Pandas, Seaborn, and Matplotlib** to uncover patterns, skewness, and correlations—particularly between glucose and hemoglobin levels using **t-SNE**.
- **Trained and compared multiple classification models** (Logistic Regression, Random Forest, SVM), with the best model (Logistic Regression) achieving **93.42% accuracy and 0.80 recall for diabetic classification**.

Housing Price Predictor

[Housing Regression Model](#)

- Implemented a **Random Forest Regression model** to analyze and predict California housing prices, identifying the top 5 most influential features impacting housing price
- **Achieved an R^2 score of 0.8153 and an average cross-validation score of 0.8156**, showing strong model generalization and performance
- Utilized Jupyter Notebook, Pandas, NumPy, scikit-learn, and Matplotlib for data cleaning, model development, feature importance extraction, and results visualization

Technologies

Languages: C, Java, Python, SQL, JavaScript, OCaml, R

Libraries/Tools: Pandas, NumPy, Matplotlib, scikit-learn, Tensor , PyTorch, Microsoft Excel

Frameworks and Environments: React, PyTorch, TensorFlow, VS Code